

Simplifying Expressions with Square Roots (Part I) Homework

Simplify.

① $\sqrt{9}$

② $\sqrt{81}$

③ $\sqrt{100}$

Use a calculator to approximate the following roots. Round to the nearest thousandth.

④ $\sqrt{5}$

⑤ $\sqrt{11}$

⑥ $\sqrt{156}$

Simplify.

⑦ $\sqrt{\frac{49}{64}}$

⑧ $\sqrt{\frac{1}{9}}$

⑨ $\sqrt{\frac{25}{4}}$

⑩ $\sqrt{\frac{16}{81}}$

Answers

3

9

10

2.236

3.317

12.49

$\frac{7}{8}$

$\frac{1}{3}$

$\frac{5}{2}$

$\frac{4}{9}$

Simplify the roots by factoring out perfect squares.

⑪ $\sqrt{12}$

⑫ $\sqrt{18}$

⑬ $\sqrt{27}$

⑭ $\sqrt{56}$

⑮ $\sqrt{60}$

Simplify.

⑯ $2\sqrt{5} - 8\sqrt{5}$

⑰ $-6\sqrt{13} + 2\sqrt{13} - \sqrt{13}$

⑱ $9\sqrt{2} - 8\sqrt{6} + 2\sqrt{2} + 5\sqrt{6}$

⑲ $-\sqrt{7} + 3\sqrt{7} - 6\sqrt{10} - \sqrt{10}$

Simplify the expressions by factoring out perfect squares and then combine like terms.

⑳ $7\sqrt{5} + \sqrt{45}$

㉑ $3\sqrt{8} - 2\sqrt{2}$

㉒ $-6\sqrt{40} + 2\sqrt{90}$

㉓ $2\sqrt{63} - 8\sqrt{7} - \sqrt{28}$

㉔ $-2\sqrt{54} - 4\sqrt{24} - 10\sqrt{96}$

$2\sqrt{3}$

$3\sqrt{2}$

$3\sqrt{3}$

$2\sqrt{14}$

$2\sqrt{15}$

$-6\sqrt{5}$

$-5\sqrt{13}$

$11\sqrt{2} - 3\sqrt{6}$

$2\sqrt{7} - 7\sqrt{10}$

$10\sqrt{5}$

$4\sqrt{2}$

$-6\sqrt{10}$

$-4\sqrt{7}$

$-54\sqrt{6}$

$$(25) 6\sqrt{112} + 2\sqrt{175} - 3\sqrt{7}$$

$$31\sqrt{7}$$

$$(26) \sqrt{75} - 7\sqrt{20} + 10\sqrt{5} + 10\sqrt{27}$$

$$35\sqrt{3} - 4\sqrt{5}$$

$$(27) -\sqrt{48} + \sqrt{160} + \sqrt{360} + 8\sqrt{3}$$

$$4\sqrt{3} + 10\sqrt{10}$$

$$(28) 5\sqrt{2} - \sqrt{2} + 2\sqrt{150} - 15\sqrt{6}$$

$$4\sqrt{2} - 5\sqrt{6}$$